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Applied Catalysis B: Environment and Energy

Date: May 2026

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Heterostructured nano-catalysts with efficient metal-oxide interfaces unlock high-performance direct methanol protonic ceramic fuel cells

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Outline

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Highlights

- A heterostructured catalyst with metal-oxide interfaces is developed.
- The catalyst exhibits good catalytic activity and coking tolerance in hydrocarbons.
- The cell demonstrates a peak power density of 1.11 Wcm⁻² on wet methanol at 700 °C.
- The cells can be operated stably on wet methanol for 100 h.
- The reaction mechanism on the catalysts is elucidated via density functional theory.

Abstract

Direct methanol protonic ceramic fuel cells (PCFCs) are attractive due to their low cost, convenient storage, and high volumetric energy density, as well as their suitability for transportation. However, the poor coking tolerance of conventional nickel-based anodes leads to their susceptibility to severe carbon deposition and significant deactivation after long-term exposure to hydrocarbons. Herein, we report a nano-catalyst of $\text{Ce}_{0.6}\text{Ni}_{0.2}\text{Cu}_{0.2}\text{O}_2$ with a heterogeneous structure that is spontaneously reduced into a $\text{Ce}_{0.6}\text{Ni}_{0.2-x}\text{Cu}_{0.2-x}\text{O}_{2-\delta}$ (CeNCO) oxide framework interfaced with a nano NiCu alloy (denoted as NC/CeNCO) under operating conditions, as confirmed by analyses of X-ray diffraction, X-ray photoelectron spectroscopy, scanning electron microscopy, and transmission electron microscopy. A Ni-BaCe_{0.7}Y_{0.06}Yb_{0.06}Zr_{0.06}Hf_{0.06}Gd_{0.06}O_{3-δ} anode-supported PCFC employing the NC/CeNCO metal-oxide catalyst achieved a peak power density of 1.11 W cm⁻² and operational stability of about 100 h at 700 °C when fueled by 35 % CH₃OH-15 % H₂O-50 % N₂. The enhanced performance and coking resistance are attributed to the efficient interfaces of Ni, Cu, and ceria-based oxide in NC/CeNCO for CH₃OH reforming, as confirmed by analyses of electrochemical performance and Raman spectroscopy with density functional theory calculations, revealing that these interfaces can enhance CH₃OH activation and promote efficient OH-mediated carbon removal via COH intermediates.

Graphical Abstract

4. Conclusions

CRedit authorship contribution statement

Declaration of Competing Interest

Acknowledgements

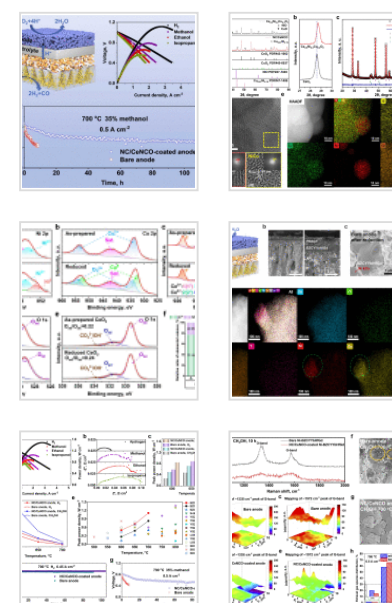
Appendix A. Supplementary material

Data Availability

References

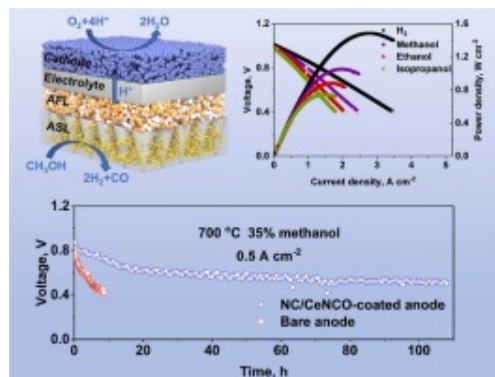
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Keywords

Protonic ceramic fuel cells; Nano-heterostructure; Catalyst; Liquid hydrocarbon fuels; Anode

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1. Introduction

With the rapid advancement of human society and scientific technology, energy and environmental issues have become increasingly prominent, necessitating the urgent identification of clean, pollution-free, new energy sources. Solid oxide fuel cells (SOFCs) have garnered significant attention due to their direct electrochemical conversion of fuels to electricity, fuel flexibility, high electrical efficiency, and ultra-low emissions [1], [2], [3], [4]. Hydrogen stands as an exceptionally promising fuel option for SOFCs. Unlike the combustion of conventional fossil fuels, which generates greenhouse gas emissions, the utilization of H₂ in SOFCs generates only water as the sole byproduct released into the

atmosphere [5]. However, the majority of hydrogen production currently relies on the reforming of hydrocarbons. The commercial application of H_2 faces challenges, including safety concerns related to its broad flammability range (4.0–74.0%) and difficulties in storage and transportation, particularly due to its extremely low liquefaction temperature of $-253\text{ }^{\circ}\text{C}$ [6]. Liquid hydrocarbon fuels (e.g., methanol, ethanol) offer advantages over hydrogen fuel, including higher volumetric energy density, lower cost, and ease of storage and transportation. Furthermore, their molecular structures are relatively simple compared to other gaseous hydrocarbons, making their application in SOFCs a promising development area [7], [8]. Protonic ceramic fuel cells (PCFCs) are well known for their good performance and excellent fuel utilization efficiency at low to medium temperatures (typically from $500\text{ }^{\circ}\text{C}$ to $700\text{ }^{\circ}\text{C}$) [9], [10], [11], [12], [13]. However, there is limited existing literature on PCFCs directly utilizing liquid hydrocarbon fuels [14], [15], with the majority of research concentrated on oxygen-ion conducting SOFCs [8], [16], [17], [18]. This is because, compared to oxygen-ion-conducting SOFCs, the anode gas components in PCFCs fueled by liquid hydrocarbons are more prone to carbon deposition [14]. The coking would block the active reaction sites, causing significant performance degradation and thereby greatly hindering the widespread application in fuel cells [19], [20], [21]. Therefore, effective prevention of carbon deposition is crucial for the development of PCFCs using liquid hydrocarbon fuels.

Various strategies have been applied to mitigate the coking issue. For example, introducing oxidants (H_2O , CO_2 , or O_2) into hydrocarbon fuels can increase the oxygen-to-carbon ratio at the anode, reducing the propensity for coking [22], [23], [24].

Incorporating nanocatalysts onto the anode to improve hydrocarbon decomposition activity and coking tolerance has also been widely studied [25]. A prevalent approach involves applying highly active reforming catalysts (e.g., $Ru-CeO_2$ and $Rh-Al_2O_3$ [26], [27], [28]) to the anode surface, significantly enhancing both power generation and carbon deposition resistance in fuel cells. However, the thermal expansion coefficient mismatch between the catalyst layer and the anode substrate could induce interfacial delamination, compromising fuel cell integrity [29]. Furthermore, the limited charge transport

capability of the catalytic layer often results in significant current-collecting challenges [30]. Another strategy is to load reforming catalysts on the inner wall of the anode holes. For example, Zhu et al. demonstrated stable cell operation for approximately 50 h with humidified methane fuel after incorporating $\text{Sm}_{0.2}\text{Ce}_{0.8}\text{O}_{1.9}$ (SDC) particles into the Ni-SDC anode through impregnation [31]; Zhang et al. impregnated Cu– CeO_2 nanocatalysts on Ni-YSZ anode, which can be operated stably for 45 h at 800 °C on methanol [32]. However, the tortuous pores of conventional anodes are not conducive to the impregnation process, limiting the amount of catalyst that can be loaded and often resulting in non-uniform deposition on the outermost part of the anode, which could severely impede gas diffusion [33]. To solve these problems, it is crucial to develop anodes with highly efficient catalysts and optimized materials/microstructures.

Studies demonstrate that nano-sized metal/oxide catalysts enhance anode resistance to carbon deposition [7], [34], [35]. Cu-based catalysts are widely used in methanol decomposition/reforming processes due to their excellent selectivity and cost-effectiveness [36], [37]. Furthermore, Cu has demonstrated a certain inertness toward C-C bond formation, suppressing carbon deposition [38]. However, Cu is also a poor catalyst for breaking the C-H bond [32]. The catalytic activity of Cu can be significantly enhanced by forming bimetallic alloys with more catalytically active metals [21]. Ni, which offers high activity at a lower cost compared to noble metals, presents a viable option [39]. Ni is one of the most active non-precious metals for breaking C-H bonds in hydrocarbons such as methane and methanol, and for catalyzing steam reforming reactions [40], [41]. Furthermore, it has been indicated that a 1:1 nickel-to-copper ratio typically suppresses coking more effectively than pure nickel [39]. On the other hand, CeO_2 -based materials are often used as substrates for anodic catalysts due to their unique oxygen ion transport capability and electrocatalytic activity, which are effective for coking removal [42]. Therefore, in this study, we rationally designed a catalyst with the composition of $\text{Ce}_{0.6}\text{Ni}_{0.2}\text{Cu}_{0.2}\text{O}_2$. Through reduction in H_2 , the catalyst undergoes an in-situ reconstruction, forming a $\text{Ce}_{0.6}\text{Ni}_{0.2-x}\text{Cu}_{0.2-x}\text{O}_{2-\delta}$ oxide framework embedded with NiCu alloy nanoparticles, denoted as NC/CeNCO. Microstructural and XPS analysis suggested

that the formation of this heterogeneous structure demonstrated efficient metal-oxide interfaces with a considerable amount of oxygen vacancies. Analyses of TEM and Raman spectroscopy showed that the exsolved Cu and Ni nanoparticles formed highly reactive NiCu alloys on ceria-based oxide, promoting the reforming of liquid hydrocarbon fuels while inhibiting carbon deposition. When applied to a Ni-BaCe_{0.7}Y_{0.06}Yb_{0.06}Zr_{0.06}Hf_{0.06}Gd_{0.06}O_{3-δ} (BZCYYbHfGd)-based anode-supported PCFCs [43], the cell with the NC/CeNCO metal-oxide catalyst achieved a peak power density of 1.11 W cm⁻² and operational stability of over 108 h at 700 °C in a fuel atmosphere of 35 % CH₃OH-15 % H₂O-50 % N₂. The improved performance and stability are attributed to the NC/CeNCO metal-oxide catalyst, which enhances CH₃OH activation and promotes efficient OH-mediated carbon removal via COH intermediates, as suggested by density functional theory calculations.

2. Experimental

2.1. Preparation of the catalyst solution

A 0.1 mol L⁻¹ Ce_{0.6}Ni_{0.2}Cu_{0.2}O₂ catalyst solution was prepared by dissolving stoichiometric amounts of Ce(NO₃)₃·6 H₂O, Ni(NO₃)₂·6 H₂O and Cu(NO₃)₂·3 H₂O in deionized water. Using a microliter syringe, 150 μL (with a loading amount of ~2.51 mg cm⁻²) of the Ce_{0.6}Ni_{0.2}Cu_{0.2}O₂ solution was infiltrated into the porous anode. It was noteworthy that each time, 10 μL of the Ce_{0.6}Ni_{0.2}Cu_{0.2}O₂ solution was infiltrated into the anode, followed by a drying process. Following deposition, the impregnated cell was calcined at 700 °C for 1 h to form Ce_{0.6}Ni_{0.2}Cu_{0.2}O₂. The preparation and infiltration methods for the Ni_{0.5}Cu_{0.5}O₂ and CeO₂ catalysts were identical to those used for Ce_{0.6}Ni_{0.2}Cu_{0.2}O₂.

2.2. Cell fabrication

This work employed an anode-supported dense thin-film electrolyte structure for the PCFC. Ni-BaCe_{0.7}Y_{0.06}Yb_{0.06}Zr_{0.06}Hf_{0.06}Gd_{0.06}O_{3-δ} (BZCYYbHfGd) cermet is applied as the anode. A high-entropy BZCYYbHfGd and PrBa_{0.5}Sr_{0.5}Co_{1.5}Fe_{0.5}O_{5+δ} (PBSCF) are used as the

electrolyte, and the cathode, respectively. High-entropy electrolytes are synthesized via solid-state reaction methods, with the specific preparation method detailed in our previous work [43]. The 60 wt% NiO/BZCYYbHfGd substrate with a unique finger-like porous structure was prepared by a phase inversion process. First, the graphite slurry was tape-cast on smooth glass, and the anode slurry was then tape-cast immediately on top. For the graphite support layer and the anode support layer, the height of the blades was adjusted to 0.35 mm and 0.90 mm, respectively. The thin layers obtained from the tape-casting were immediately immersed in deionized water for 10 h to complete the phase inversion process. The layers were removed and dried in ambient air overnight. The dried layers were stamped into pellets (~ 15 mm in diameter) and subjected to a preheating process (1000°C for 2 h) to remove the organic components and the graphite layer. A thin film of the anode functional layer and the BZCYYbHfGd electrolyte was drop-coated onto the substrate and sintered at 1475°C for 5 h. The PBSCF cathode was deposited onto the BZCYYbHfGd electrolyte layer via screen-printing and sintered in air at 950°C for 2 h (area of 0.196 cm^2). Experimental details can be found in the [Supporting Information](#).

2.3. Electrochemical testing

Individual cells were supported by anodes that were adhered to ceramic tubes using ceramic adhesive and heated in a tube furnace to the test temperature. Water was added to methanol to achieve a methanol-to-water molar ratio of 7:3 for vapor reforming. During the initial stage, hydrogen was introduced into the anode at a flow rate of 30 mL min^{-1} for 30 min at the operating temperature to reduce the nickel oxide. Subsequently, the temperature of the vaporizer was raised above 140°C . An aqueous methanol solution was then fed into the vaporizer using a peristaltic pump at a rate of $86\text{ }\mu\text{L min}^{-1}$. The resulting mixture of methanol vapor and water vapor was delivered to the anode using N_2 at a flow rate of 30 mL min^{-1} as the carrier gas. The cathode was exposed to ambient air. The air electrode (cathode) was connected to a silver wire using silver paste for current collection. A schematic of the electrochemical test setup for a protonic ceramic fuel cell (PCFC) operating on liquid hydrocarbon fuels such as methanol is provided in [Figure S1](#). Electrochemical characterization, including current density-

voltage curves and electrochemical impedance spectra (EIS), was measured between 600 and 700 °C using a PARSTAT MC 200 system. The exhaust was analyzed through a dry tube using a gas chromatograph (Shimadzu GC-2014). Liquid hydrocarbon fuels, such as methanol, were tested under an atmosphere composition of 35% CH₃OH - 15% H₂O - 50% N₂.

2.4. Computational methods

Density functional theory (DFT) computations were conducted with the Vienna *Ab initio* Simulation Package (VASP) [44], [45] to investigate the electronic and catalytic effects of NiCu nanoparticles supported on the substrate and bare Ni(111) surfaces for methanol decomposition and carbon removal. The projector augmented-wave (PAW) method [46] and the Perdew–Burke–Ernzerhof (PBE) functional within the generalized gradient approximation (GGA) [47] were employed, with a plane-wave cutoff energy of 415 eV. To account for the strong on-site Coulomb interactions in Ce 4f states, a Hubbard U correction ($U_{\text{eff}} = 5.0$ eV) [48], [49], [50] was applied using the Dudarev approach [51]. Transition states were located using the climbing image nudged elastic band (CI-NEB) method [52]. We performed spin-polarized periodic density functional theory (DFT) calculations to identify feasible reaction pathways for methanol (CH₃OH) decomposition and to investigate efficient surface carbon removal, aiming to elucidate the exceptional carbon tolerance of the nickel-based anode under methanol conditions using a Ce_{0.6}Ni_{0.2-x}Cu_{0.2-x}O_{2-δ} (CeNCO) oxide framework coated with a nano-NiCu alloy (denoted as NC/CeNCO). All DFT calculations were performed using the Vienna *ab initio* Simulation Package (VASP) [44], [45] with the projector-augmented wave (PAW) method [46] and the Perdew–Burke–Ernzerhof (PBE) generalized gradient approximation (GGA) functional [47], employing a plane-wave cutoff energy of 415 eV. A Γ -centered 6×6×6 k-point mesh was used for Brillouin zone (BZ) sampling, and the Gaussian smearing method ($\sigma = 0.05$ eV) was applied in the optimization of bulk CeO₂ and NiCu-doped CeO₂ (Ce_{0.5}Ni_{0.25}Cu_{0.25}O₂) structures using the PBE + U approach with an effective U value of U_{eff} of 5.0 eV for Ce, similar to previous studies [48], [49], [50]. A vacuum layer of 15 Å was introduced along the surface normal for two-dimensional slab calculations to prevent interactions between

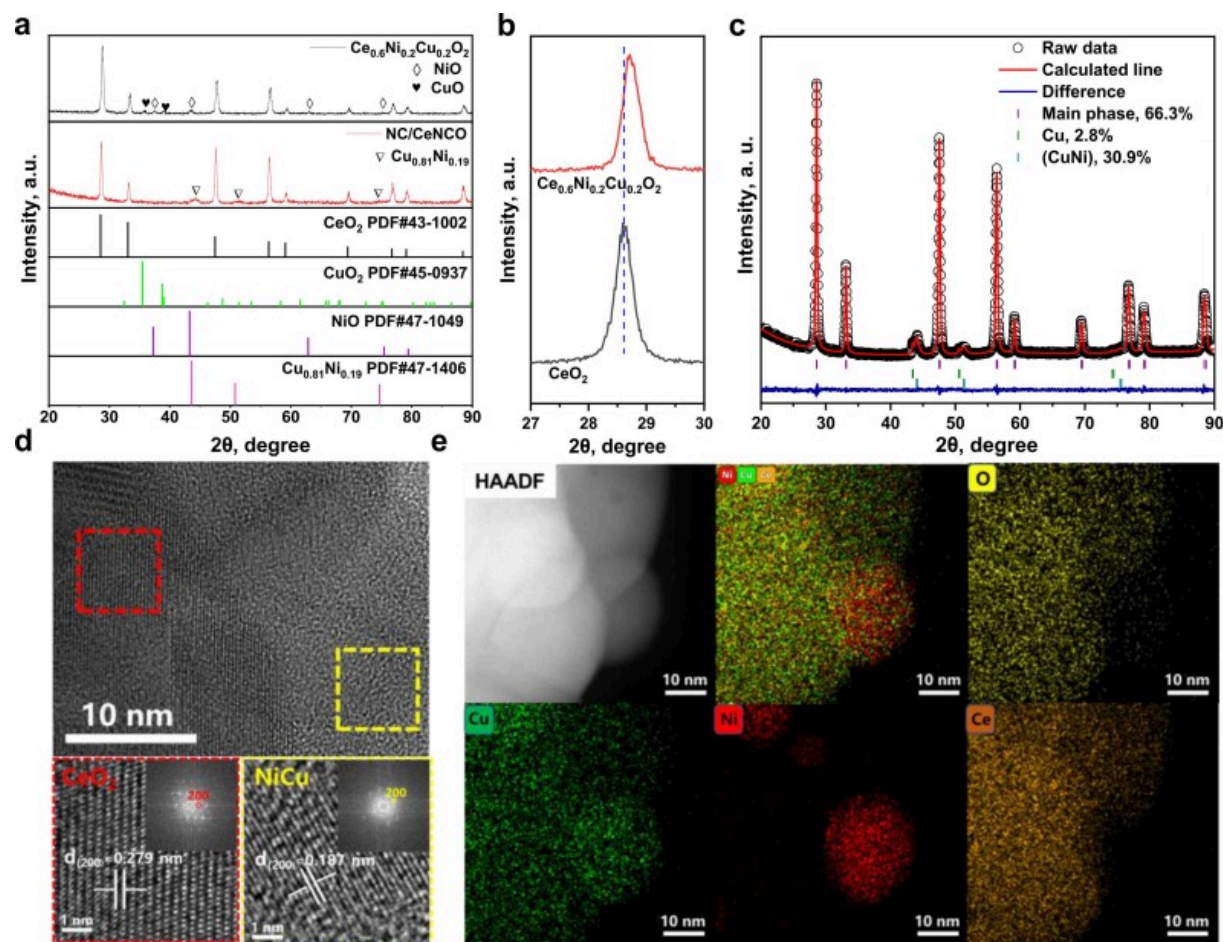
periodic images of the supercells, using a Γ -point-only k -point mesh. A dipole correction was applied to eliminate errors arising from slab asymmetry. The most stable surface among the three low-index planes was selected based on previous studies (e.g., Ni(111)) [7], [53] to identify the catalytically active sites for mechanistic investigations of methanol decomposition and carbon removal. Based on our previous study, a three-layer $p(3 \times 6)$ Ni(111) slab model (54 Ni atoms) was constructed, and a CeNCO row (7 Ce, 5 Ni, 4 Cu, and 16 O atoms) was deposited on its surface, representing the CeNCO/Ni(111) configuration. While the bottom two layers were constrained to the bulk lattice parameters, the top layer was allowed to relax along with the adsorbates (intermediate species and CeNCO). Gas-phase molecules were optimized in a cubic simulation cell with dimensions of $15 \times 15 \times 15 \text{ \AA}^3$. The adsorption energy (E_{ads}) of an adsorbate is calculated as $E_{\text{ads}} = E(\text{adsorbate-surface}) - E(\text{surface}) - E(\text{adsorbate})$, where $E(\text{adsorbate-surface})$, $E(\text{surface})$, and $E(\text{adsorbate})$ are the DFT total energies of the adsorbed system, the clean surface, and the isolated adsorbate in the gas phase, respectively. Using six images for each CI-NEB calculation, the climbing-image nudged elastic band (CI-NEB) method [52] was employed to identify transition states and map the minimum energy pathways for methanol decomposition and carbon removal. Visualization of three-dimensional structures and charge density difference analyses were performed using the VESTA software [54].

3. Results and discussion

3.1. Microanalyses of catalysts

The $\text{Ce}_{0.6}\text{Ni}_{0.2}\text{Cu}_{0.2}\text{O}_2$ catalyst powders were successfully synthesized via the sol-gel method at 800°C for 2 h. Fig. 1a presents a comparison of the XRD patterns of the $\text{Ce}_{0.6}\text{Ni}_{0.2}\text{Cu}_{0.2}\text{O}_2$ powders before and after 2 h of reduction in a 97% H_2 -3% H_2O atmosphere. The XRD pattern before reduction distinctly displayed peaks corresponding to NiO (PDF#47-1049) and CuO (PDF#45-0937). Furthermore, the magnified XRD patterns of $\text{Ce}_{0.6}\text{Ni}_{0.2}\text{Cu}_{0.2}\text{O}_2$ and CeO_2 powders (Fig. 1b and S2, Supporting Information)

revealed a significant shift to the right in the peak position of $\text{Ce}_{0.6}\text{Ni}_{0.2}\text{Cu}_{0.2}\text{O}_2$ near 28° relative to CeO_2 . This shift was attributed to the smaller ionic radii of Ni^{2+} (0.69 Å) and Cu^{2+} (0.73 Å) compared to Ce^{4+} (0.87 Å). These findings indicate the successful doping of Ni and Cu into the CeO_2 lattice. The disappearance of NiO and CuO peaks and the appearance of peaks corresponding to $\text{Ni}_{0.19}\text{Cu}_{0.81}$ alloy can be seen in the XRD pattern of powders after the reduction. The structural parameters of NC/CeNCO powders were further analyzed using the Rietveld refinement ([Fig. 1c](#)). The results indicated that the NC/CeNCO demonstrated a CeO_2 oxide phase (66.3%) with a space group of Fm-3m and lattice parameters of $a = b = c = 5.4126 \text{ Å}$ ($R_{\text{wp}} = 2.94\%$, $R_p = 2.23\%$, $\chi^2 = 1.193$), alongside two exsolved metallic phases: Cu (2.8%) and NiCu alloy (30.9%). The residual metal content (Ni + Cu) within the CeO_2 lattice was approximately 6.3% (Comprehensive data are provided in [Table S1](#), [Supporting Information](#), and [Supplementary Note 2](#)).



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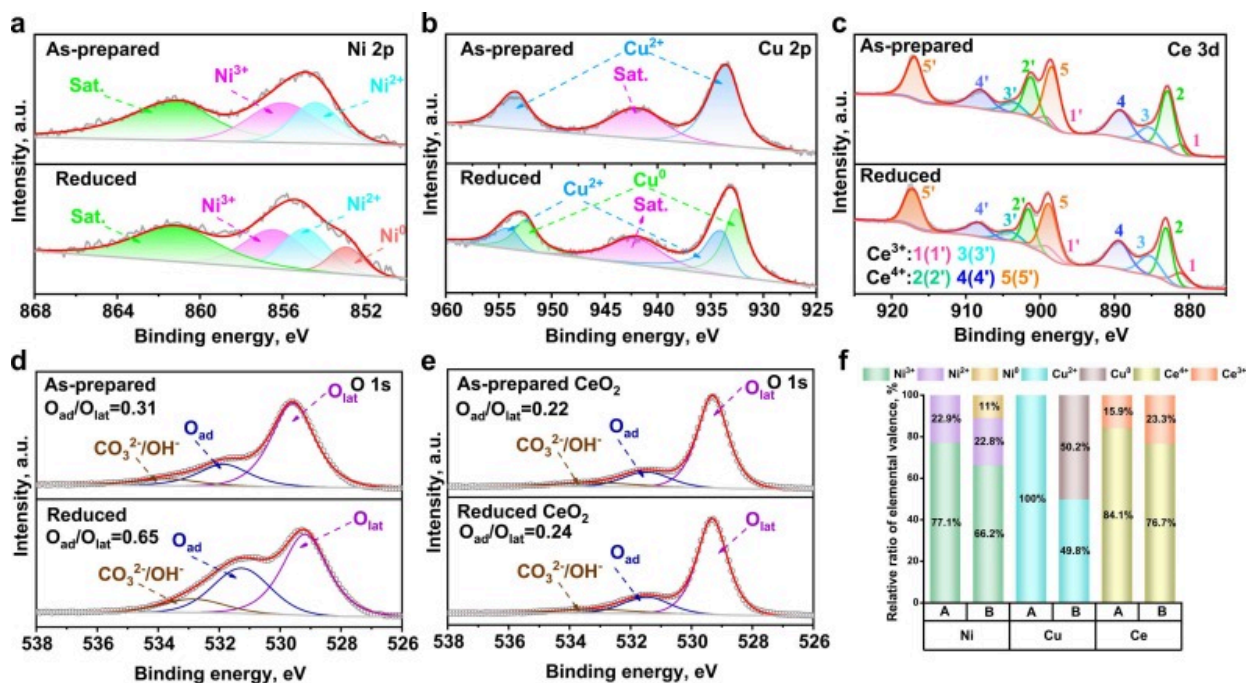
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Fig. 1. Phase and structure characterizations of the catalyst. **(a)** XRD patterns of $\text{Ce}_{0.6}\text{Ni}_{0.2}\text{Cu}_{0.2}\text{O}_2$ before and after reduction; **(b)** Enlarged XRD patterns of $\text{Ce}_{0.6}\text{Ni}_{0.2}\text{Cu}_{0.2}\text{O}_2$ and CeO_2 ; **(c)** Rietveld refinement of the reduced $\text{Ce}_{0.6}\text{Ni}_{0.2}\text{Cu}_{0.2}\text{O}_2$ (NC/CeNCO) at 700 °C in H_2 for 2 h; **(d)** HRTEM image of NC/CeNCO grains with insets displaying fast Fourier transform (FFT) patterns: red box (CeO_2 substrate) and yellow box (NiCu alloy nanoparticle); **(e)** Corresponding HAADF-STEM image and EDS elemental mapping.

As shown in [Fig. 1d](#), high-resolution transmission electron microscopy (HRTEM) imaging reveals two distinct nanoparticle types dispersed on the NC/CeNCO substrate. [Fig. 1d \(red box\)](#) displays the HRTEM image of the substrate, where the measured 0.279 nm lattice spacing matches the (200) plane of CeO_2 . The yellow box in [Fig. 1d](#) shows a magnified image of the nanoparticles. The measured 0.187 nm lattice spacing matches the (200) plane of NiCu alloy, confirming the formation of NiCu alloy phases in the NC/CeNCO powders, consistent with XRD analysis. [Fig. 1e](#) presents the high-angle annular dark-field (HAADF) image and corresponding EDS mapping of the NC/CeNCO, revealing localized Ni/Cu segregation alongside homogeneous Ce/O distribution. The tendency of the Ni and Cu elements to stay together to form a polymetallic phase further confirms the generation of a nanoscale NiCu alloy. Furthermore, EDS mapping ([Figure S3, Supporting Information](#)) revealed a broader Cu segregation region compared to Ni, suggesting the presence of minor metallic Cu alongside the predominant NiCu alloy phase, consistent with XRD Rietveld refinement results.

[Fig. 2](#) compares the XPS spectra of Ce 3d, Cu 2p, Ni 2p, and O 1s of the $\text{Ce}_{0.6}\text{Ni}_{0.2}\text{Cu}_{0.2}\text{O}_2$ powders before and after the reduction. [Fig. 2a](#) shows that Ni in $\text{Ce}_{0.6}\text{Ni}_{0.2}\text{Cu}_{0.2}\text{O}_2$ exists in a mixture of +2 and +3 valence states, whereas the three chemical states of nickel appeared after reduction (Ni^0 , Ni^{2+} , and Ni^{3+}) are 10.94 %, 22.83 % and 66.23 %, respectively [\[6\]](#). Similarly, Cu in $\text{Ce}_{0.6}\text{Ni}_{0.2}\text{Cu}_{0.2}\text{O}_2$ is mainly present as Cu^{2+} before reduction, while after reduction it is present in two chemical states, Cu^{2+} and Cu^0 , with percentages of 49.77 % and 50.23 %, respectively ([Fig. 2b](#)) [\[6\]](#). The simultaneous presence of Ni^0 and Cu^0 in the reduced samples further suggests that Ni and Cu partially exsolved into the metallic state. This conclusion is in agreement with the preceding XRD and TEM characterizations. The Ce 3d XPS spectrum in [Fig. 2c](#) confirms the partial reduction of Ce^{4+} to Ce^{3+} . The reduction is followed by a significant increase in Ce^{3+} ([Fig. 2f](#)), and the presence of Ce^{3+} correlates with increased oxygen vacancy density, enhancing methanol decomposition activity [\[7\]](#). The evolution of O species was analyzed for both $\text{Ce}_{0.6}\text{Ni}_{0.2}\text{Cu}_{0.2}\text{O}_2$ and CeO_2 before and after reduction. Typically, the molar ratio of adsorbed oxygen to lattice oxygen represents the oxygen vacancy content [\[55\]](#).

Specifically, a larger value of O_{ad}/O_{lat} indicates a higher concentration of oxygen vacancies. As shown in Fig. 2d, e, and S4 (Supporting Information), NC/CeNCO exhibits the highest O_{ad}/O_{lat} ratio, suggesting the highest oxygen vacancy concentration. These oxygen vacancies may act as active sites that promote efficient water adsorption and dissociation, thereby continuously generating abundant surface OH^* species [24]. This significantly enhances the catalytic activity of carbon removal reactions during hydrocarbon conversion [56], [57], a mechanism further corroborated by the DFT calculations presented in the following section.



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Fig. 2. XPS analysis of the catalyst powder. XPS spectra of $Ce_{0.6}Ni_{0.2}Cu_{0.2}O_2$ before and after reduction at 700 °C in H_2 : (a) Ni 2p; (b) Cu 2p; (c) Ce 3d; (d) O 1s; (e) O 1s XPS spectra of CeO_2 before and after reduction in H_2 at 700 °C; (f) Summary of valence

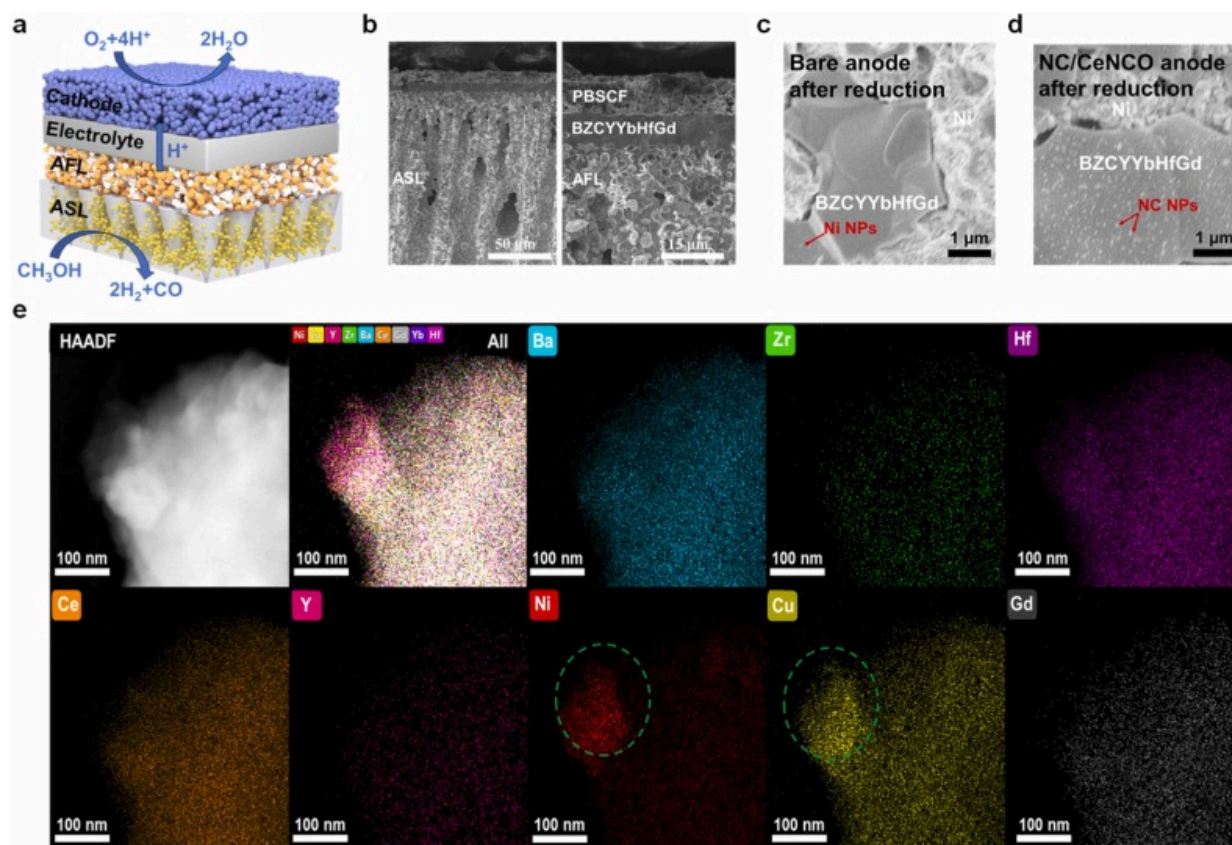
changes before and after $\text{Ce}_{0.6}\text{Ni}_{0.2}\text{Cu}_{0.2}\text{O}_2$ reduction, where A represents the as-prepared powder and B represents the reduced powder.

The results of XRD, XPS, and TEM analyses indicated not only the partial exsolution of Ni and Cu elements to form NiCu alloy nanoparticles, but also a decrease in Ce valence state and an increase in oxygen vacancy concentration on the reduced $\text{Ce}_{0.6}\text{Ni}_{0.2}\text{Cu}_{0.2}\text{O}_2$ surface. Therefore, for simplicity, the reduced catalyst was denoted as NC/CeNCO. When employed as the anode catalyst in PCFCs, these exsolved metals in NC/CeNCO were expected to form a highly active interface, modified by multi-metallic NiCu nanoparticles on the Ni-based anode surface, inhibiting carbon deposition while promoting methanol decomposition.

3.2. Single-cell architectures and the electrochemical performance

As schematically shown in Fig. 3a, in a typical fuel cell test, vaporized liquid fuel enters the anode along with nitrogen gas, which subsequently decomposes into carbon monoxide (CO) and hydrogen (H_2) within the anode's modified layer [7]. Due to the presence of finger holes, the syngas (CO and H_2) could quickly reach the anode active sites near the anode/electrolyte interface to electrochemically react and produce electrical energy. As shown in Fig. 3b, the cell consists of a porous $\text{PrBa}_{0.5}\text{Sr}_{0.5}\text{Co}_{1.5}\text{Fe}_{0.5}\text{O}_{5+\delta}$ (PBSCF) cathode layer, a dense BZCYYbHfGd high-entropy electrolyte layer, a porous Ni-BZCYYbHfGd anode functional layer (AFL), and a finger-like porous Ni-BZCYYbHfGd anode support layer (ASL). Detailed SEM images of both reduced bare and NC/CeNCO-coated anodes in 3 % H_2O humidified hydrogen are presented in Fig. 3c and d. Fig. 3c shows Ni nanoparticles (NPs) exsolution on the BZCYYbHfGd surface in the bare anode cell. In contrast, in the cell with NC/CeNCO-coated anode, many catalyst NPs are uniformly coated on the anode surface, exhibiting higher surface coverage than the exsolved Ni particles in the bare anode (Fig. 3d). To unambiguously identify the composition of the exsolved NPs, we simulated the test conditions by grinding the cells loaded with NC/CeNCO catalysts into powders for TEM analyses after reduction under wet H_2 for 2 h (Fig. 3e). The distribution of Ba, Ce, Zr, Y, Hf, Gd, Ni, and Cu elements shows that Ni and Cu tend to assemble to form NiCu alloys. The well-dispersed NiCu NPs on the anode surface

could inhibit the aggregation of Ni while protecting Ni from coking [39]. In the case of methanol, for example, the NC/CeNCO NP catalysts coated on the inner surface of the porous anode promote methanol reforming while inhibiting carbon deposition.

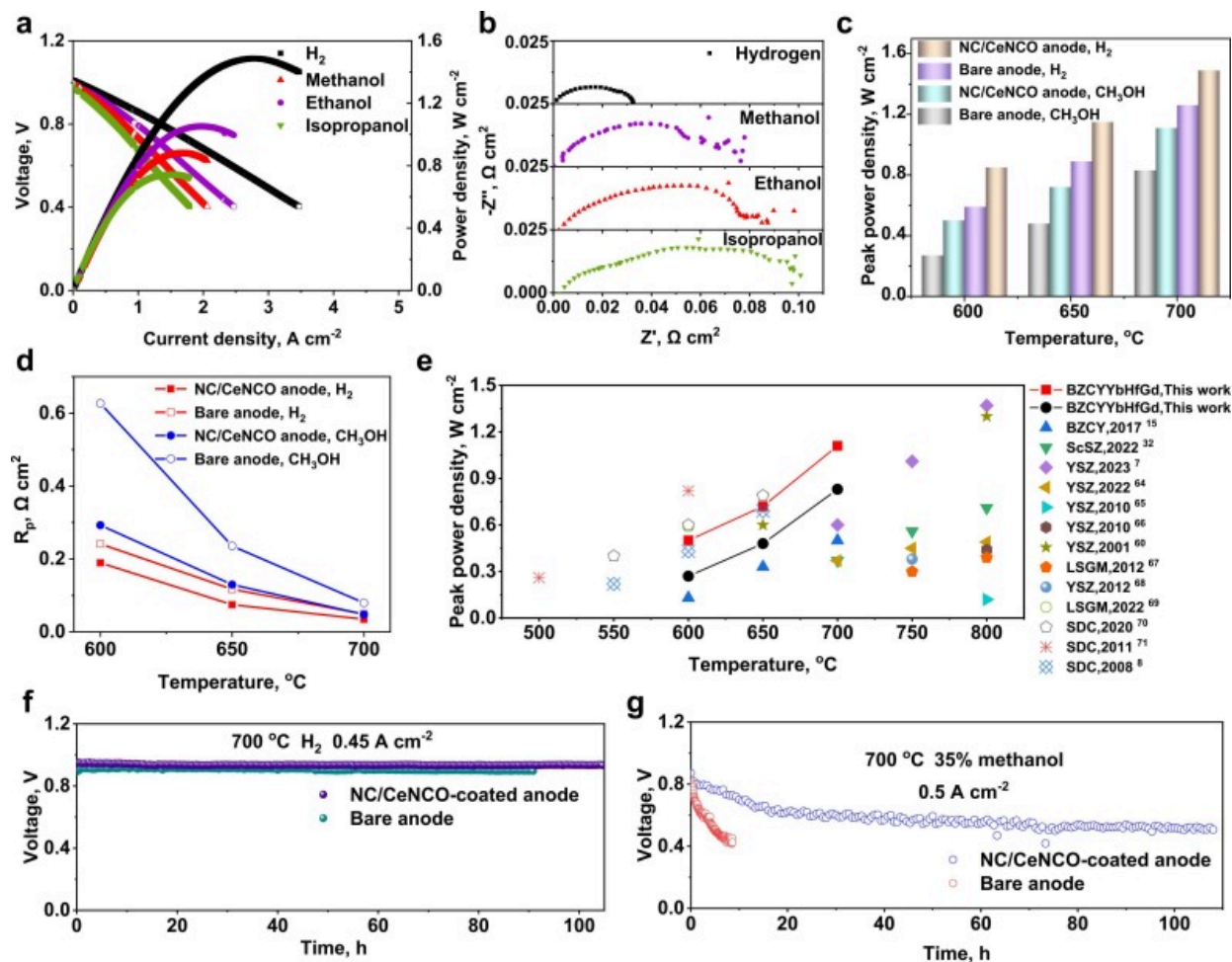


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Fig. 3. Structure of a PCFC with the NC/CeNCO-coated anode. **(a)** Schematic of the methanol-fueled single cell employing NC/CeNCO as anode coating; **(b)** a cross-sectional SEM image of the cell (left); A detailed SEM image of the cell (right), including a porous PBSCF cathode, a dense BZCYYbHfGd electrolyte ($\approx 10 \mu m$), and a porous Ni-BZCYYbHfGd anode; **(c)** Detailed SEM image of the bare and **(d)** NC/CeNCO-coated anode; **(e)** TEM mapping of cells with NC/CeNCO-coated anodes reduced in wet H_2 at $700^\circ C$ for 2 h.

The IV and electrochemical impedance spectra (EIS) curves were collected at 700 °C for the single cells with the NC/CeNCO catalyst using wet hydrogen, 35 % wet methanol, 35 % wet ethanol, and 35 % wet isopropanol as fuels ([Figure S1](#)) to assess the catalyst's reforming performance. As shown in [Fig. 4a](#), when wet H₂ was supplied to the PCFC, the peak power density (PPD) of the cell with the NC/CeNCO-catalyst was 1.48 W cm⁻² at 700 °C. When other liquid hydrocarbons were used as fuels, a significant decrease in PPD was observed, with the values of 1.05, 0.87, and 0.74 W cm⁻² when using fuels of methanol, ethanol, and isopropanol, respectively, at 700 °C. Furthermore, the open-circuit voltage of individual cells varies when using different liquid fuels. This variation arises from differences in the fuel's molecular structure and the complexity of the involved reactions, which affect the actual effective hydrogen partial pressure at the anode. Larger, more complex molecules are more prone to generating diluent or unreactive by-products (e.g., CH₄, C₂H₄) during reforming, thereby lowering the OCV [\[58\]](#), [\[59\]](#). As shown in [Fig. 4b](#), the cell exhibited relatively low polarization resistance (R_p) under hydrogen conditions. However, when liquid hydrocarbon fuels were used, the R_p increased significantly, especially when isopropanol was used. There are several possible reasons for this phenomenon. First, the carbon and hydrogen contents in different fuels are different, potentially leading to variations in open-circuit voltages when operated in a single cell. Second, the energies required to break the C-C and C-H bonds also vary, depending on the capabilities of the catalyst and the inherent nature of the fuel. Third, differences in the thermodynamic or kinetic nature of the reforming reaction may also localize these variations [\[60\]](#), [\[61\]](#), [\[62\]](#). As a result of the above crucial factors, the cells exhibited different polarization behaviors, leading to varied power outputs.



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Fig. 4. Electrochemical performance and durability of the PCFC with the NC/CeNCO catalyst in CH₃OH and other liquid fuels. **(a)** I-V-P curves and **(b)** corresponding EIS curves of NC/CeNCO-coated single cells at 700 °C using wet hydrogen, 35 % wet methanol, 35 % wet ethanol, and 35 % wet isopropanol as the fuel; **(c)** Peak power density comparison (bare and NC/CeNCO-coated anodes) at 600–700 °C using wet H₂/35 % wet CH₃OH; **(d)** Polarization resistances (R_p) comparison (bare and NC/CeNCO-coated anodes) at 600–

700 °C using wet H₂/35 % wet CH₃OH; **(e)** Performance benchmarking against state-of-the-art methanol-fueled SOFCs at 700 °C; **(f)** Long-term stability of both bare and NC/CeNCO-coated anode cells at 700 °C using wet H₂ (under the current density of 0.45 A cm⁻²) or **(g)** 35 % wet CH₃OH (under the current density of 0.5 A cm⁻²). The composition of the fuel injected into the anode is 35 % CH₃OH-15 % H₂O-50 % N₂.

Fig. 4c summarizes the PPD of the cells with and without catalysts at various temperatures and fuel atmospheres (detailed I-V-P curves are shown in [Figure S6, Supporting Information](#)). The NC/CeNCO-coated anode cell in wet CH₃OH exhibited PPD values of 0.50, 0.72, and 1.11 W cm⁻² at 600, 650, and 700 °C, respectively, lower than the PPD values in wet H₂ (0.85, 1.15, and 1.49 W cm⁻² at 600, 650, and 700 °C, respectively). The performance differences may be attributed to the lower electrochemical reaction kinetics of CH₃OH fuel compared to H₂ [63], leading to an increase in ohmic resistance (R_{ohm}) and polarization resistance (R_p) ([Figure S7, Supporting Information](#)). Furthermore, the NC/CeNCO-coated anode achieved higher peak power densities (PPDs) than the bare anode in both humidified H₂ and CH₃OH at 600–700 °C, demonstrating enhanced electro-oxidation of H₂ and reforming activity of CH₃OH. The performance variation of the cells primarily originated from the difference in R_p contributed by the anodes and cathodes ([Fig. 4d, Figures S7 and S8, Supporting Information](#)). The NC/CeNCO-coated anode exhibited lower R_p than the bare anode at 600–700 °C due to enhanced catalytic activity ([Fig. 4d](#)). **Fig. 4e** shows a detailed comparison of the PPD of our cell with other cells reported in the literature when using methanol fuel [7], [8], [15], [32], [60], [64], [65], [66], [67], [68], [69], [70], [71]. The NC/CeNCO-coated cells rank among the top performers and demonstrate good durability in cells with similar structures. Detailed information on these fuel cells is given in [Table S2 \(Supporting Information\)](#). For example, at 700 °C in 35 % wet CH₃OH, our work obtained a PPD of 1.11 W cm⁻², which was higher than that reported for a PCFC based on a BaZr_{0.3}Ce_{0.5}Y_{0.2}O_{3-δ} (BZCY) electrolyte and a Ni-BZCY cermet anode (0.50 W cm⁻² at 700 °C while using 68 % CH₃OH-32 % N₂ as fuel) [15], demonstrating the decent catalytic activity of the NC/CeNCO-coated anode for CH₃OH decomposition. Dang et al. reported that the PCFC exhibited good durability, operating

stably for 117 h at 700 °C in wet CH₃OH; however, the cell's performance (IV and EIS curves) in wet methanol was not provided [14]. Furthermore, our performance is also better than the SOFCs with Ru/CR_{5-x}O coated porous anodes and YSZ electrolytes (a P_{max} of 0.60 W cm⁻² at 700 °C in CH₃OH, the highest documented power density for CH₃OH-SOFCs with oxygen-ion-conducting YSZ electrolytes) [7].

Fig. 4f compares the cell voltages of bare and NC/CeNCO-coated Ni-BZCYYbHfGd anodes at a current density of 0.45 A cm⁻² and 700 °C in wet H₂ (3 % H₂O). Both single cells demonstrated excellent stability in humidified H₂ for 85 or 100 h, with no significant performance degradation observed regardless of catalyst coating. The operational durability of cells directly fueled by CH₃OH was more critical for practical applications than that of those fueled by H₂. **Fig. 4g** shows that during 108 h of testing at 700 °C and 0.5 A cm⁻², the voltage of the cell with NC/CeNCO-coated anode slowly decreased during the first 20 h, after which it was gradually stabilized. The gradual performance decay likely results from slight carbon deposition on the Ni anode, originating from residual methanol decomposition. In contrast, the voltage of the bare anode cell without catalyst modification decreased rapidly from 0.8 V to 0.4 V within 10 h due to rapid coking on the Ni-based anode. The open-circuit voltage (OCV) is also an important indicator for evaluating the stability of the cells. The OCVs of both cells were tested in methanol and hydrogen atmospheres for 2 h. As shown in **Figure S9 (Supporting Information)**, the bare anode cell showed a slight decrease in open-circuit voltage (OCV) under CH₃OH as compared to the NC/CeNCO-coated anode, although both returned to the original 1 V when switched to the H₂ atmosphere. The substantial increase in stability demonstrates that NC/CeNCO catalysts can greatly inhibit coking formation on the Ni-BZCYYbHfGd anode, extending the life of the single cell in liquid hydrocarbon fuels. This was confirmed by SEM images of the cells after stability testing in wet methanol (**Figure S10, Supporting Information**). Significant carbon deposition was observed in the bare anode pores, clogging the pores and hindering fuel reforming and diffusion, leading to rapid and irreversible cell degradation [72]; this was not the case in the NC/CeNCO-coated anode pores. Interestingly, the NC/CeNCO-coated anode demonstrated the recombination of a

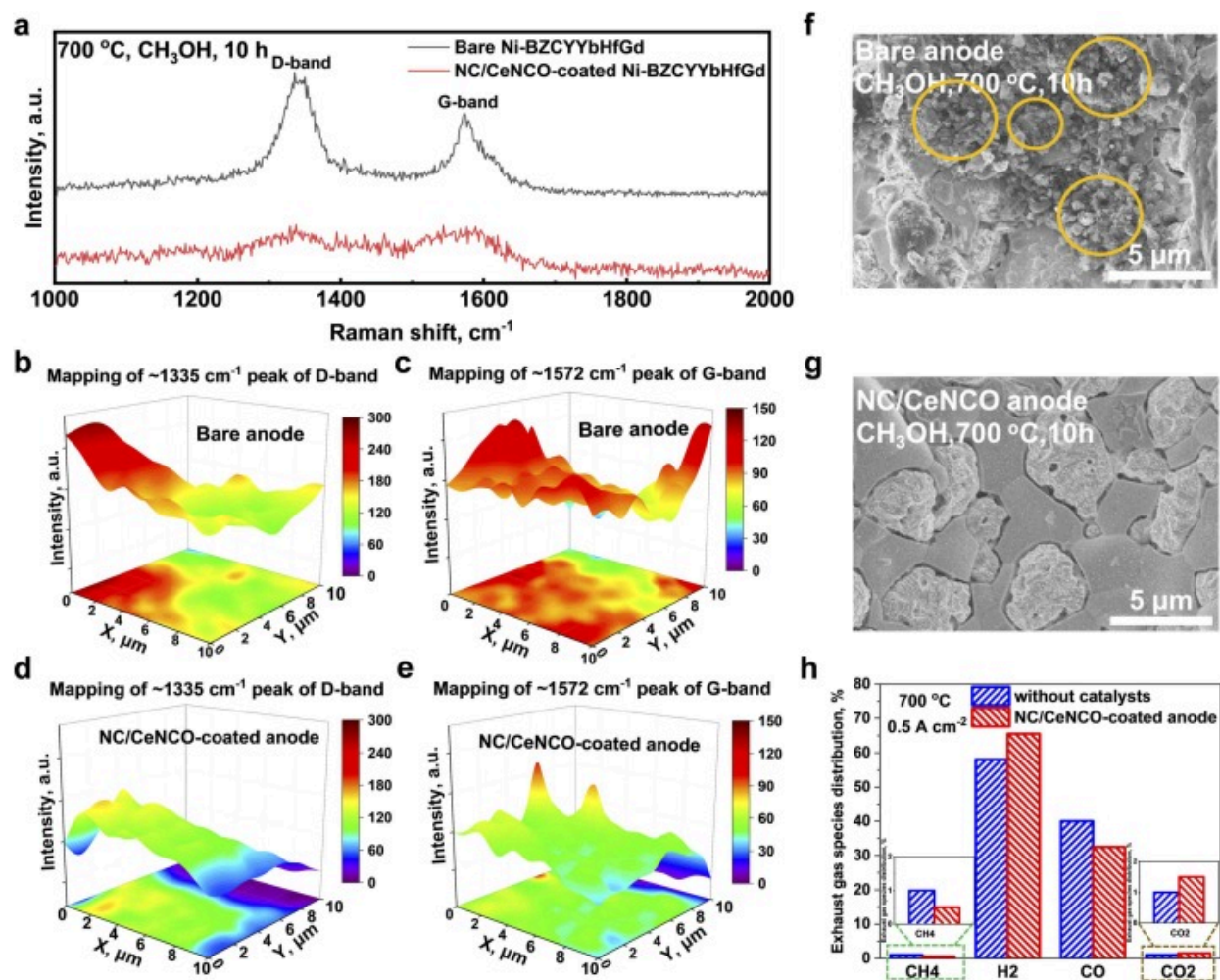
thin film and well-defined NPs on the surface of Ni-BZCYYbHfGd grains ([Figure S11, Supporting Information](#)). This heterogeneous anode may be key to efficiently decomposing methanol and inhibiting carbon deposition.

For better comparison, fuel cells with the anodes coated with CeO₂ or NiCu alone were also tested at 700 °C ([Figure S12, Supporting Information](#)). The results showed that all three fuel cells (NC/CeNCO, CeO₂, and NiCu) exhibited improved performance compared to bare anode-based cells, both in wet H₂ and wet CH₃OH. The NC/CeNCO-coated cells showed the highest PPD and lowest R_p when operated with either methanol or hydrogen fuel, demonstrating enhanced catalytic performance. To better explore the synergistic effects of CeO₂ and NiCu, the durability of fuel cells with CeO₂ and NiCu-coated anodes in wet methanol was also tested ([Figure S13, Supporting Information](#)). The stability of two cells (CeO₂ and NiCu-coated anodes) was improved relative to that of bare anodes. This may be due to the strong oxygen storage and migration capability of reduced cerium dioxide, which improved the coking tolerance [73], while Cu enrichment on the Ni surface in NiCu alloys could block the active sites for hydrocarbon fuel dissociation, thus improving the coking resistance [39]. However, the NC/CeNCO-coated cells still showed the most stable operation at the test current condition of 0.5 A cm⁻² ([Fig. 4g](#)). These test results suggest that the synergistic effect of Ce with Ni and Cu elements may be responsible for the excellent output power density and good durability.

3.3. Surface Raman mapping

SEM analysis and Raman mapping further confirmed the excellent resistance against carbon deposition of NC/CeNCO-coated Ni-BZCYYbHfGd anodes. [Fig. 5a](#) shows typical Raman spectra obtained from bare and NC/CeNCO-coated anodes after exposure to 35 % wet CH₃OH at 700 °C for 10 h under OCV conditions, which represents a more severe condition for coking [7]. Distinct carbon signatures were detected on the two anodes, with characteristic Raman peaks at 1335 cm⁻¹ (D-band) and 1572 cm⁻¹ (G-band). The D-band (disordered carbon, lacking long-range symmetry) showed reversible removal during regeneration, whereas the G-band corresponded to in-plane hexagonal carbon

vibrations [74], [75]. Compared to the bare anode, only very weak D and G bands were observed in the spectra of the NC/CeNCO-coated anode, indicating that carbon deposition was greatly suppressed. Raman mapping ($10\ \mu\text{m} \times 10\ \mu\text{m}$) of carbon characteristic peaks (D-band: $1335\ \text{cm}^{-1}$, G-band: $1572\ \text{cm}^{-1}$) revealed significantly more intense coke deposition on the bare anode compared to the NC/CeNCO-modified surface, particularly evident in the G-band intensity (Fig. 5b-e). This was also confirmed by SEM images of the anode surface after testing (Fig. 5f and g). Fig. 5f shows significant morphological changes in the bare anode after wet methanol exposure, with numerous surface-deposited particles likely resulting from carbon coking. In comparison, the NC/CeNCO-coated anode exhibited minimal carbon deposition (Fig. 5a and g), demonstrating significantly enhanced coking resistance. Fig. 5h shows the exhaust gas composition analysis from the PCFC operating on 35 % CH_3OH -15 % H_2O -50 % N_2 fuel, with H_2 and CO as the main gas components. Hydrogen was produced by methanol dehydrogenation ($\text{CH}_3\text{OH} \rightarrow \text{CO} + 2\ \text{H}_2$) and the steam reforming reaction ($\text{CH}_3\text{OH} + \text{H}_2\text{O} \rightarrow \text{CO}_2 + 3\ \text{H}_2$) [14]. The proportion of H_2 was significantly increased by the introduction of the catalyst compared to the bare anode fuel cell, and this improvement could be attributed to the high catalytic activity of the NC/CeNCO catalyst in the methanol decomposition and steam reforming reactions. Furthermore, the decrease in the ratio of CO to CH_4 and the slight increase in the CO_2 ratio demonstrate enhanced fuel oxidation efficiency with the catalyst. Exhaust analysis confirmed the NC/CeNCO nanocatalyst boosted methanol decomposition efficiency in PCFCs, enhancing the performance and stability.



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Fig. 5. Evaluation of coking resistance. (a) Raman spectra of bare and NC/CeNCO-coated anodes after 10 h operation with 35 % wet CH₃OH at 700 °C; Corresponding Raman mappings of carbon signatures (D-band: ~1335 cm⁻¹, G-band: ~1572 cm⁻¹) for (b, c) bare and (d, e) NC/CeNCO-coated anodes after 10 h operation under identical conditions; SEM characterization of (f) bare and (g) NC/CeNCO-coated anodes following 10 h operation under 35 % wet CH₃OH at 700 °C; (h) anode effluent gas composition analysis at 0.5 A

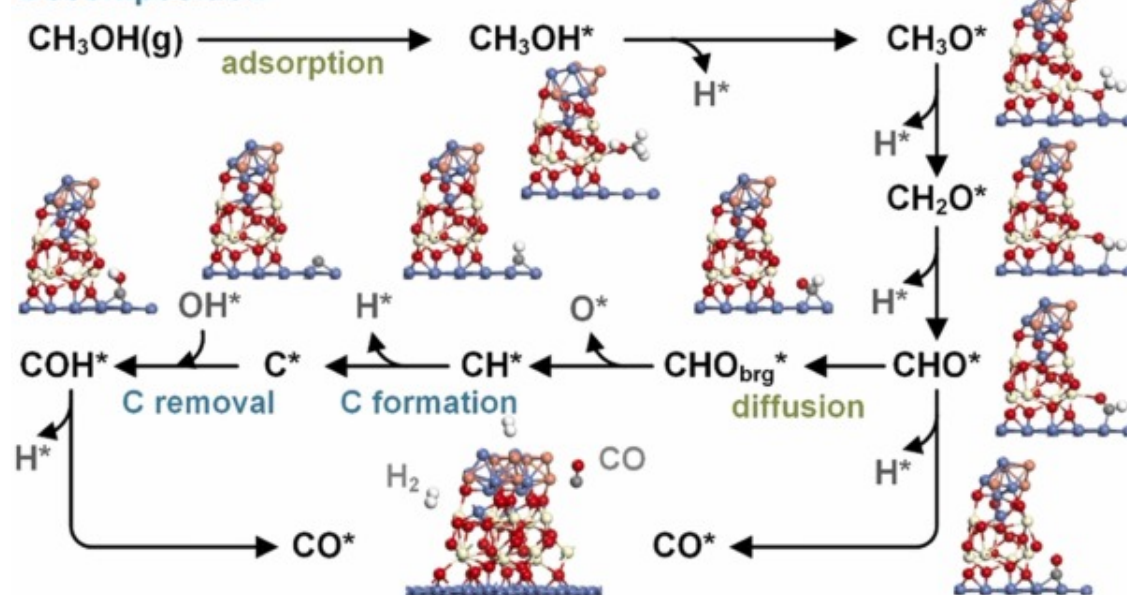
cm^{-2} constant current density; the insets show enlarged views of the CH_4 and CO_2 distributions.

3.4. Theoretical investigation of methanol decomposition pathways

Further, we employed DFT calculations to investigate the elementary steps involved in elucidating the experimental findings in the extraordinary coke tolerance of CeNCO/Ni anodes during methanol oxidation (Fig. 6), similar to our previous study [7]. As depicted in Figure S14, the bulk structures of Ni and NiCu-doped CeO_2 were optimized to construct slab models (Figures S15 and S16). As reported [76], coke formation on Ni-based anodes is usually mitigated by maintaining high steam-to-carbon ratios. However, consistent with our earlier findings [7], the present system operates effectively at a low steam-to-carbon ratio (0.43), suggesting that additional carbon-management mechanisms are active. Similarly, we examined the critical intermediate species associated with nanostructured CeNCO on Ni anodes and compared them with those observed on traditional Ni anodes. In this study, only the dehydrogenation of the O-H bond was considered according to the previous studies [53], [77], as methanol deposition on Ni surfaces (Table S3). Mechanistic insights for pure Ni(111) are summarized in Table S4 and Figure S15. Methanol activation initiates with O-H bond cleavage, followed by sequential dehydrogenation through CH_3O^* and CH_2O^* intermediates. An asterisk (*) identifies species adsorbed on the surface. The rate-limiting step is the dehydrogenation of CH_3O^* , which requires 1.01 eV. Subsequently, carbon accumulation occurs via CH^* fragmentation (1.09 eV), and removal through OH-assisted COH^* formation is kinetically demanding (1.19 eV). Diffusion analysis further shows that OH^* migration is energetically more favorable than carbon migration, yet insufficient for rapid Ni surface cleaning. In contrast, CeNCO/Ni(111) significantly modifies the energy landscape (Table S4 and Figure S16). The barrier for the rate-limiting step ($\text{CH}_3\text{O}^* \rightarrow \text{CH}_2\text{O}^* + \text{H}^*$) decreases to 0.35 eV, highlighting the role of CeNCO-derived oxygen species in facilitating hydrogen abstraction. Carbon removal is also significantly accelerated. The $\text{C}^* + \text{OH}^* \rightarrow \text{COH}^*$ step proceeds with an energy barrier of 0.42 eV, substantially lower than the 1.19 eV observed on bare Ni. This enhancement is attributed to the strong CeNCO-Ni interfacial interaction. According to transition-state theory [78],

reducing the activation energy for the rate-determining step by approximately 65 % can dramatically accelerate the methanol decomposition rate under operating conditions. The mechanistic analysis reveals two primary advantages of CeNCO/Ni, including enhanced CH_3OH activation and efficient OH-mediated carbon removal via COH intermediates. These processes align with an internal steam reforming pathway enabled by CeNCO's strong water affinity and redox flexibility. These attributes deliver accelerated methanol decomposition and robust coke mitigation, enabling CeNCO/Ni to surpass conventional Ni-based anodes under comparable fuel cell conditions.

Decomposition



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Fig. 6. Proposed reaction sequence for enhancing methanol decomposition and coking resistance via a COH intermediate species on CeNCO/Ni(111). An asterisk denotes surface species.

4. Conclusions

In summary, an NC/CeNCO heterostructured nano-catalyst with efficient metal-oxide interfaces was designed and fabricated to enhance the electrochemical performance and stability of direct methanol protonic ceramic fuel cells (PCFCs). The catalyst-modified cell achieved 1.11 W cm^{-2} peak power density at 700°C with stable operation for approximately 100 h using 35 % wet methanol as the fuel. The remarkable power output and long-term cell stability could be attributed to the NC/CeNCO catalyst coating, providing more active sites for the electrochemical reaction. Furthermore, a significant amount of the exsolved NiCu alloy nanoparticles were anchored on the substrate, demonstrating high catalytic activity in methanol dehydrogenation and steam reforming reactions. DFT calculations also support that the formed catalyst can enhance CH_3OH activation, and the coking can be removed efficiently via COH intermediates. The NC/CeNCO nanocatalysts incorporated in our PCFCs effectively promote methanol decomposition while preserving the structural integrity of the Ni-based anode, demonstrating a promising approach for developing high-performance direct methanol PCFCs.

CRediT authorship contribution statement

Kang Xu: Formal analysis, Data curation. **Fan He:** Validation. **Yangsen Xu:** Formal analysis. **Yuhe Liao:** Formal analysis. **Kotaro Sasaki:** Formal analysis, Data curation. **Xirui Zhang:** Formal analysis. **Feng Zhu:** Formal analysis. **Xuan Yang:** Writing – original draft, Validation, Formal analysis. **Liyan Chen:** Formal analysis, Data curation. **YongMan Choi:** Writing – review & editing. **Ying Liu:** Formal analysis. **Yu Chen:** Writing-review & editing, Supervision, Resources, Project administration.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was financially supported by the Introduced Innovative R&D Team of Guangdong ([2021ZT09L392](#)), the National Natural Science Foundation of China ([22179039](#)), the Pearl River Talent Recruitment Program ([2019QN01C693](#)), the Guangdong Basic and Applied Basic Research Foundation ([2024A1515010448](#)), [Guangzhou Science and Technology Project](#) ([2024A04J3079](#)), and Zijin Mining Group Co., Ltd. ([5405-ZC-2023-00008](#)). Computational studies were supported by the National Science and Technology Council (NSTC Grant No. [111-2221-E-A49-003-MY3](#) and [114-2221-E-A49-179-MY3](#)), the National Center for High-performance Computing (NCHC), and the Higher Education Sprout Project of the National Yang Ming Chiao Tung University and the Ministry of Education (MOE), Taiwan. DFT calculations were performed using the resources of the Center for Functional Nanomaterials, which is a U.S. DOE Office of Science Facility, at Brookhaven National Laboratory under Contract No. DE-SC0012704.

Appendix A. Supplementary material

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Supplementary material

Data Availability

Data will be made available on request.

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